

A smaller Quipus recorder

Editable electrical design, connection tables and a proposed physical arrangement.

2,000 mAh / replaceable protected LiPo / offline recording first

This revision is designed around two onboard microphones, a digital expansion port, one speaker, microSD and a small display. Power, Start, Stop, Up and Down are physical controls. USB-C supplies charging and programming.

Power and battery

1S LiPo, 3.7 V nominal / 4.2 V charge. JST PH battery connector plus separate NTC. Hardware power-button controller; charger remains active when recording electronics are off.

Controller and storage

ESP32-S3-WROOM-1-N16R8. Native USB, microSD in 1-bit mode, a 1.3-inch SPI display and RTC. Start has a direct wake pin; three menu buttons use an I2C expander.

Audio

Two IM69D128S PDM microphones. Keyed short-wire digital expansion. MAX98357A class-D amplifier with a separate differential speaker output. Keep microphones clear of speaker vibration.

Release status: circuit review draft. Exact PCB footprints, copper routing, native ERC/DRC and hardware qualification are not complete. This package is for design review; it must not be sent straight to fabrication.

Battery life: six hours of recording is the first measured acceptance goal. A week of 5-6 recording hours each day is not promised with 2,000 mAh. Larger packs need matching polarity, temperature limits, current capability and gauge settings.

144 circuit entries / 92 named nets / 9 editable schematic sheets

3D placement study

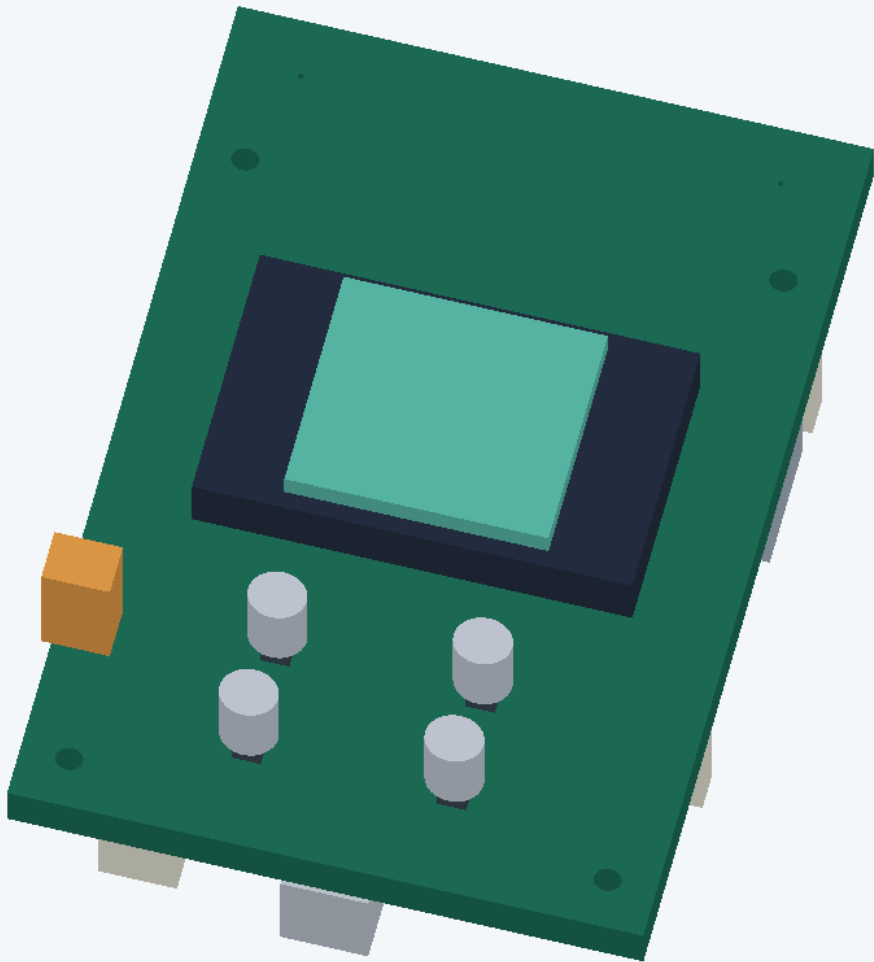
STEP assembly included. Body envelopes are provisional; this is not a model of a routed or manufactured PCB.

QUIPUS A1 / PLACEMENT STUDY

65 × 105 × 1.6 mm PCB · 2,000 mAh pack connects separately

NOT FOR MANUFACTURE

FRONT / SCREEN + SOUND OPENINGS



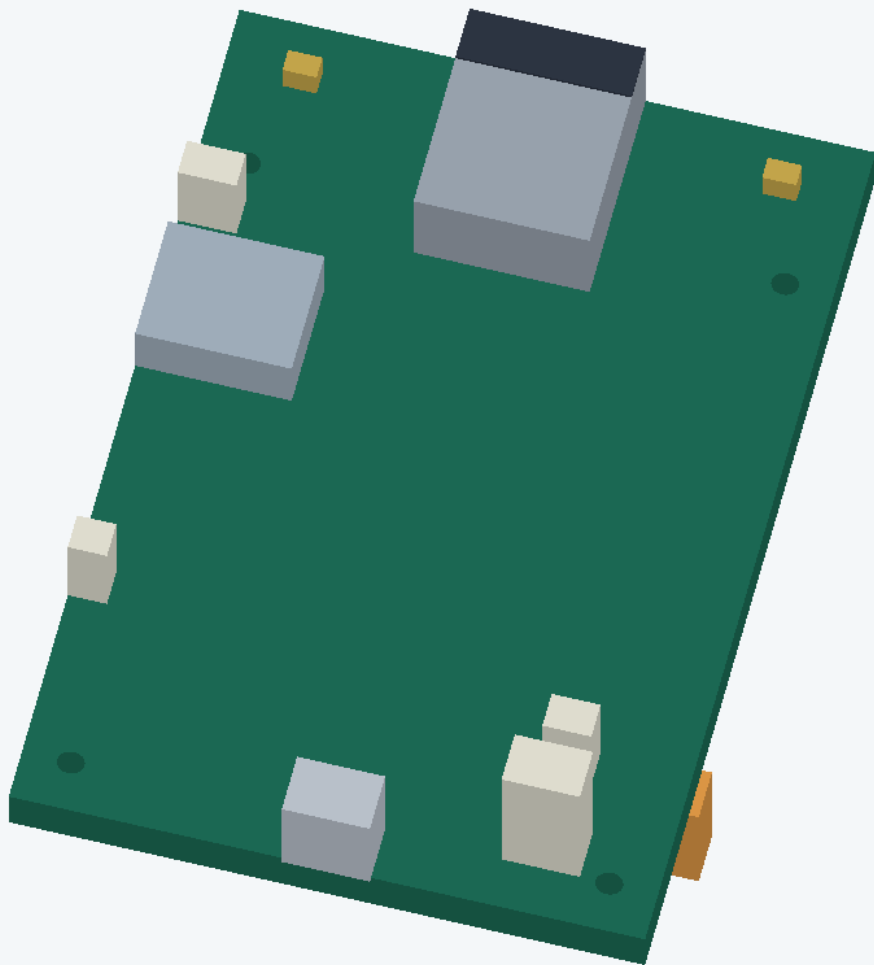
FRONT

1.3-inch display, four menu buttons; power on the left.

REAR

ESP32 antenna at top; SD exits right; USB-C exits bottom.

REAR / COMPONENT ENVELOPES



MICROPHONES

Rear-mounted; sound passes through two front PCB holes.

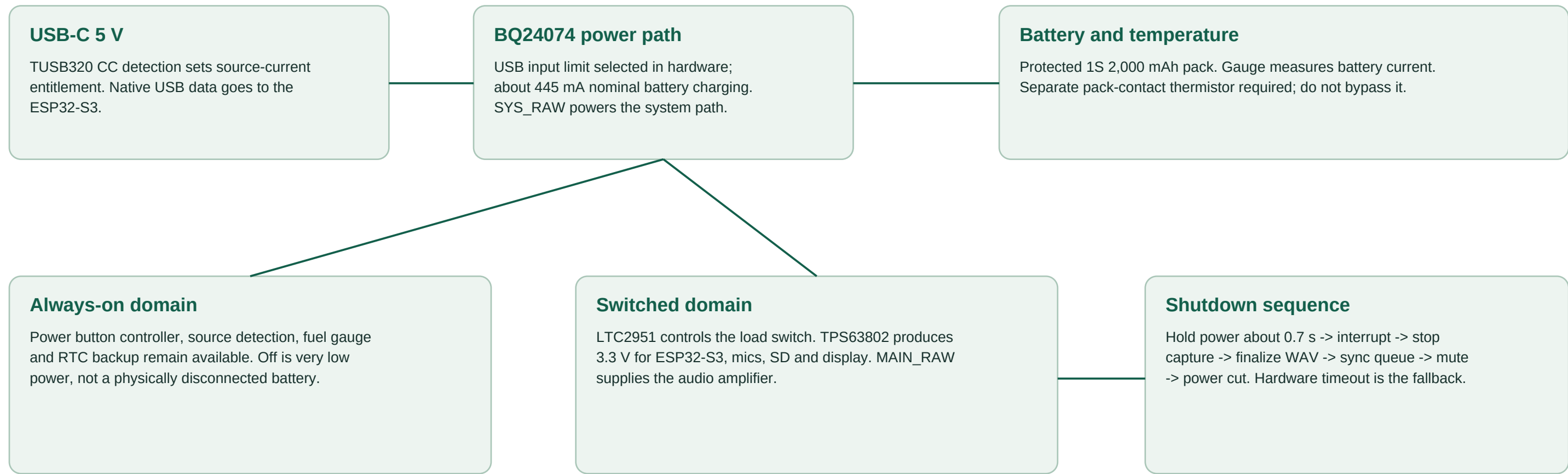
NOT MODELLED

Battery, speaker, wiring, routed copper or final component pads.

Component shapes are placement envelopes. Several heights and connector outlines are provisional; final mechanical drawings take priority.

Power architecture

Different rails serve the charger, low-power controls and recording electronics.



USB default mode is limited until the source permits more current. An ordinary USB socket and a USB-C high-current charger are not interchangeable. Off-state charging must work without the ESP32 running; USB suspend and enumeration are explicit firmware responsibilities.

The BQ24074 thermistor protection window must be checked against the purchased pack: its usual 0-50 C nominal window does not automatically satisfy a pack rated for charging only at 0-45 C. Pack/NTC selection and thermal qualification remain release blockers.

Capacity, current and storage

These are planning calculations, not observed runtime or measured audio power.

Average battery current*	Recording hours	Days at 6 h recording
80 mA	20.0	3.33
120 mA	13.33	2.22
180 mA	8.89	1.48
240 mA	6.67	1.11

*All loads and conversion losses are included in these hypothetical battery-side currents. Calculation: 2,000 mAh x 0.8 reserve allowance / average current. Idle time, uploads and voice playback reduce the working-day result.

Electrical design check	Planning value	Implication
Battery energy	3.7 V x 2 Ah = 7.4 Wh	Capacity is not output power
Week of 5-6 h/day	35-42 h recording	Requires <=38-46 mA before idle
Charger nominal current	445 mA	Capacity/current = 4.49 h, before CV tail
3.3 V rail peak allowance	0.85 A	Validate converter at lowest loaded input
Main-switch peak estimate	1.45 A at 3.0 V input / 85% efficiency	Includes 0.35 A speaker allowance; derate below this input
Stereo PCM16 at 16 kHz	64,000 bytes/s	1.3824 GB for six hours
Mono PCM16 at 16 kHz	32,000 bytes/s	0.6912 GB for six hours
Classic WAV / FAT32 limit	About 18.64 h per stereo file	Segment audio; retain one meeting manifest

Battery/connector current ratings and actual voltage drops must be checked together. SD write peaks, radio bursts and speaker playback can overlap; average current alone cannot size the power circuit.

Power component calculations

Values are derived from manufacturer limits; routing, component bias and temperature still require validation.

Circuit / setting	Nominal result	Tolerance or consequence
BQ24074 RISET = 2.00k 1%	445 mA charge	492.4 mA maximum using specified gain tolerance
BQ24074 RILIM = 1.24k 1%	1.298 A USB input	1.401 A maximum; only with >=1.5 A CC advertisement
TPS63802 FB = 511k / 91k	3.3077 V rail	3.2196 to 3.3981 V calculated tolerance band
LTC2951 OFFT = 100nF	674 ms press qualification	Button is momentary; press timing is not a power-current rating
LTC2951 KILLT = 1uF	6.54 s typical shutdown grace	Verify capacitor tolerance/leakage and SD flush worst case
BQ24074 TMR = 68k	9.07 h typical timer	6.73 to 11.45 h calculated; review after capacity change
Gauge shunt = 0.010 ohm	20 mV / 40 mW at 2 A	Use 1%, low-TCR Kelvin connections; selected resistor rated 0.25 W

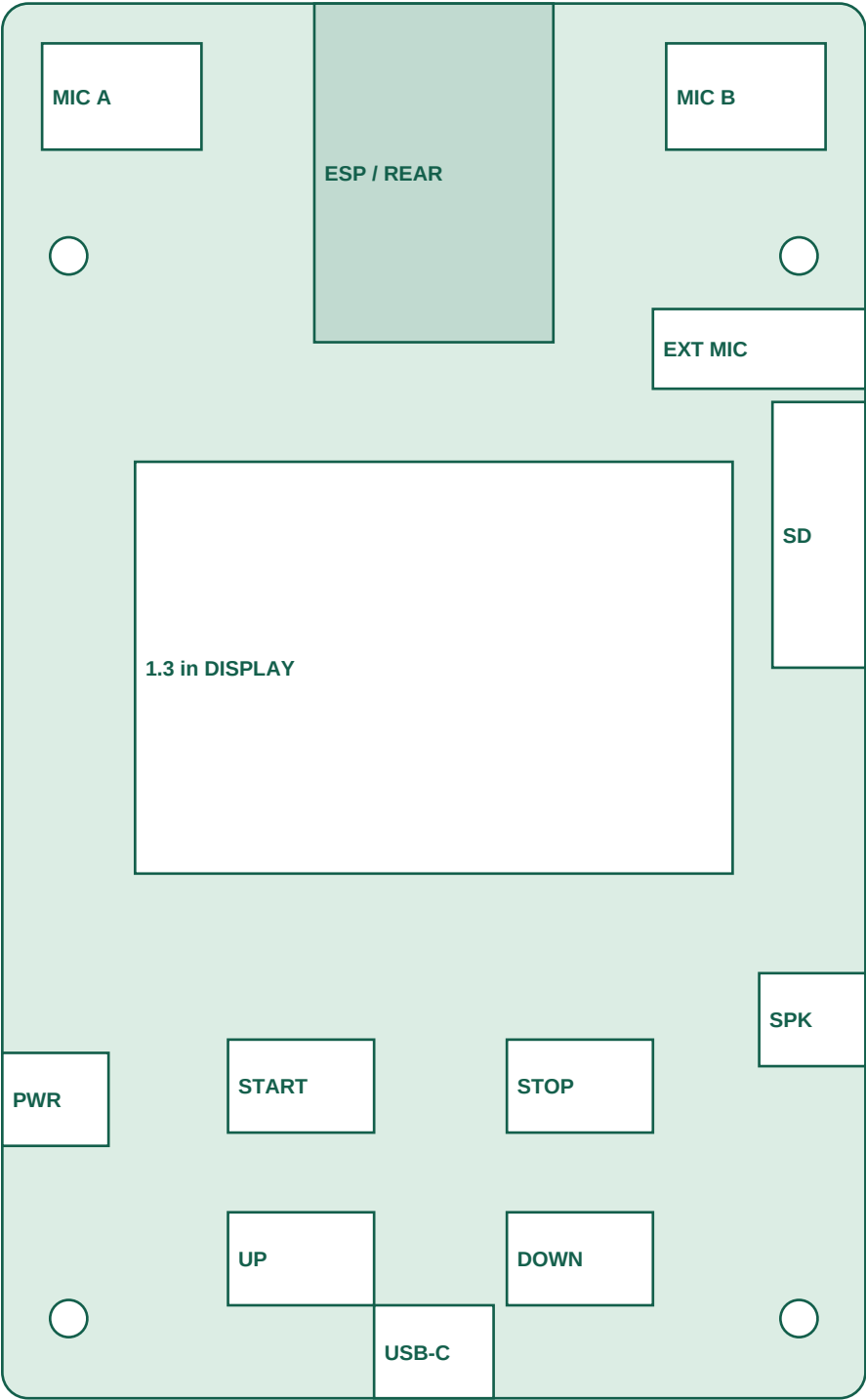
USB state	EN1 / EN2	Allowed charger input mode
Default source, not enumerated	0 / 0	100 mA mode
Default source, USB configured for 500 mA	1 / 0	500 mA mode
Type-C source advertises >=1.5 A	0 / 1	RILIM mode, about 1.30 A nominal

USB suspend requested	1 / 1	Suspend mode; verify actual total connector current
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Charge-path heating example: $(5.0 - 3.2) \text{ V} \times 0.445 \text{ A} = 0.80 \text{ W}$ before additional system-path losses. Thermal copper and enclosure temperature matter even at this moderate charge current. No thermal rise or loop stability is claimed from these calculations.

Proposed physical arrangement

65 x 105 mm placement study, not an approved board outline or a footprint drawing.



TOP / USER-FACING SIDE

Microphones face the user

Bottom-port MEMS bodies mount on the rear, with individual PCB sound holes opening toward the front/top case surface. Keep an isolated acoustic path to each mic; no foam or adhesive across the port.

Antenna clearance

ESP module antenna is at the upper edge. Preserve its manufacturer keepout on every layer and in the enclosure. No battery, metal screen frame or cabling may invade that space.

Side access

SD card must be removable from the side. Speaker is external to the PCB and sits behind a side-facing grille. Keep the speaker cavity separate from microphone inlet channels.

Battery and service

Battery sits behind the lower board with insulated clearance. Its exact size is not selected. Leave a wire route to JST battery/NTC and speaker connectors; preserve service access to BOOT/RESET.

Four-layer construction

Signal/components / continuous ground / power and slow signals / signal/components. The drawing is a zoning proposal only: final dimensions follow footprints, routing and battery fit.

Connector and harness definitions

Use the numbered contact in the selected manufacturer drawing; do not infer polarity from wire colour.

J1 / GCT USB4105-GF-A

A1_B12: GND -> GND; A4_B9: VBUS -> VBUS_5V; B8: SBU2 -> NC; A5: CC1 -> USB_CC1; B7: D-2 -> USB_DM; A6: D+1 -> USB_DP; A7: D-1 -> USB_DM; B6: D+2 -> USB_DP; A8: SBU1 -> NC; B5: CC2 -> USB_CC2; B4_A9: VBUS -> VBUS_5V; B1_A12: GND -> GND; SH1: SHIELD -> GND; SH2: SHIELD -> GND; SH3: SHIELD -> GND; SH4: SHIELD -> GND

GCT USB4105 PCB-top-mount USB2 receptacle, 0.95 mm shell stakes | Sixteen mating contacts map to twelve solder tails: A1/B12, A4/B9, B4/A9, B1/A12 share pads. Four shield stakes connect to ground. Pad names below explicitly combine shared contacts; verify the imported CAD footprint uses the same mapping. No external CC Rd resistors.

J2 / JST S2B-PH-SM4-TB(LF)(SN) battery

1: PACK+ -> BAT_PACK_P; 2: PACK- -> GND

JST PH 2mm 2-position right angle SMT | Board-defined pin 1 is positive. Verify pack polarity physically. Protected 1S 2000 mAh pack, standard 4.2 V charge, at least 2 A discharge capability. Swap with device off and USB unplugged.

J3 / 3-pin keyed NTC connector - JST SH family candidate

1: NTC -> BAT_NTC; 2: NTC_RETURN -> GND; 3: KEY_NC -> NC

1mm SH 3pin exact drawing pending | Three positions prevent swapping with the two-position speaker connector. A 10 kOhm 103AT-2 thermistor must be thermally attached to the cell. Do not populate a fixed bypass resistor.

J6 / DM3AT-SF-PEJM5

1: DAT2 -> SD_D2_PU; 2: CD_DAT3 -> SD_D3_PU; 3: CMD -> SD_CMD; 4: VDD -> 3V3_SYS; 5: CLK -> SD_CLK; 6: VSS -> GND; 7: DAT0 -> SD_D0; 8: DAT1 -> SD_D1_PU; A: CARD_DETECT_A -> SD_CD_N; B: CARD_DETECT_B -> GND; SH: SHIELD -> GND

Hirose DM3AT push-push microSD top mount | Native 1-bit SDMMC. DAT1/2/3 have 10k pull-ups even though not routed to ESP. Hirose drawing names separate detect contacts A/B, not 9/10; CAD footprint must map those named pads explicitly. Symbol SH represents all shell solder stakes; do not short detect-A to shell through a generic footprint. Detect closes A-B with card inserted. Enclosure must allow ejection travel.

Connector and harness definitions

Use the numbered contact in the selected manufacturer drawing; do not infer polarity from wire colour.

J7 / B8B-PH-K-S(LF)(SN)

1: BL -> LCD_BL; 2: RST -> LCD_RST_N; 3: DC -> LCD_DC; 4: CS -> LCD_CS_N; 5: CLK -> LCD_SCLK; 6: DIN -> LCD_MOSI; 7: GND -> GND; 8: VCC -> 3V3_SYS

JST PH 8-position 2.0mm vertical through-hole | Custom Quipus connector numbering matches Waveshare SKU15867 MODULE J2 PH2.0, not its reversed J1 header. Vendor schematic J2:1BL,2RST,3DC,4CS,5SCK,6DIN,7GND,8VCC. Match signals by continuity, do not assume supplied cable colors/orientation. External module 45x31mm includes ST7789 240x240 and NPN backlight driver; no direct LED-current drive from GPIO.

J4 / SM04B-SRSS-TB

1: 3V3 -> MIC_3V3; 2: GND -> GND; 3: CLK -> PDM_CLK_EXT; 4: DATA -> PDM_DIN1

JST SH 1.0mm 1x4 right-angle SMD | Mates SHR-04V-S with SSH-003T-P0.2-H contacts. Internal digital PDM accessory with an initial cable target of 50 mm or less; not USB, TRRS or analog. Accessory needs local mic bypass and a 100 ohm DATA series resistor. Pin assignment is Quipus-specific; verify cable orientation.

J5 / SM02B-SRSS-TB

1: SPK+ -> SPK_P; 2: SPK- -> SPK_N

JST SH 1.0mm 1x2 right-angle SMD | Mates SHR-02V-S. Neither pin is ground. Short twisted pair to 8 ohm, 2 W nominal dynamic speaker; exact speaker part pending. Speaker connector intentionally differs from battery series.

Firmware contract and validation

Circuit capability does not mean the existing firmware already supports the new board.

Capture without a network

I2S0 receives two PDM microphone slots. Start at 16 kHz PCM / 2.048 MHz PDM clock. Preserve both channels during evaluation; mixing them blindly is not beamforming. SD files and meeting IDs must survive reboot.

Avoid bus and pin conflicts

I2S1 supplies speaker playback. SPI serves the display; SD uses the dedicated 1-bit bus. GPIO35-37 are reserved for the N16R8 module PSRAM. Native USB uses GPIO19/20. Do not reuse strapping pins for normal buttons.

Power and UI

Set the gauge capacity/profile for the fitted cell; read real charger status. Keep Wi-Fi and backlight off during offline capture when possible. Only show a time-to-full estimate while charging and after current measurements are stable.

Time and shutdown

Provision RTC backup settings and disable its trickle charging. Battery removal can lose time; mark timestamps uncertain until resynchronised. Service the shutdown interrupt, close files and release power within the hardware timeout.

Measurements that decide release

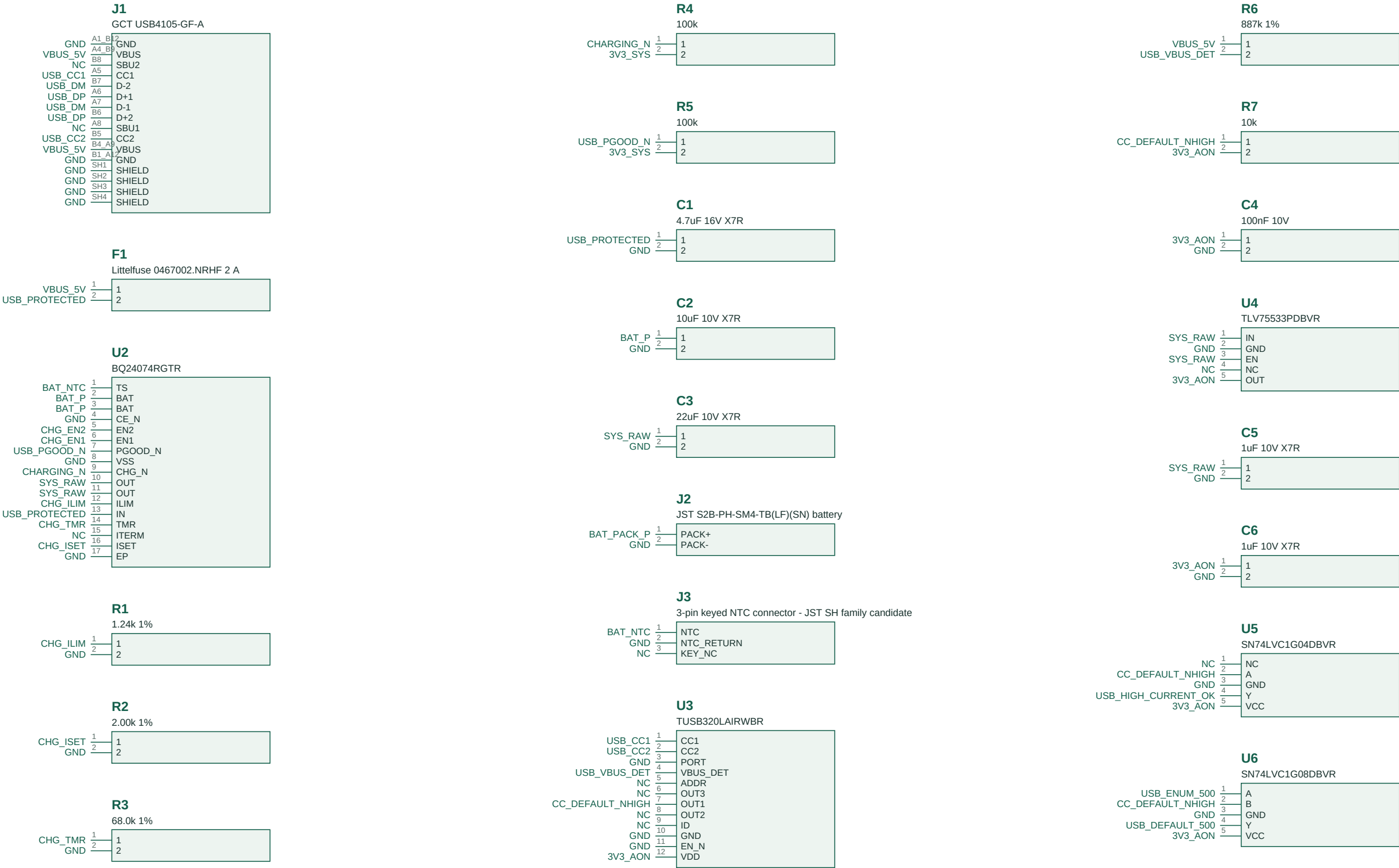
Scope power rails during simultaneous SD writes, Wi-Fi and speaker playback. Measure charging temperature, current limits, off-state current and six-hour capture. Interrupt power during file updates and validate queue recovery.

CAD checks still outstanding

Generated source is structurally checked. Native KiCad/ERC and EasyEDA import have not been run here. Exact footprints, routing, DRC, acoustic stackup, USB signal integrity and prototype approval are required before fabrication.

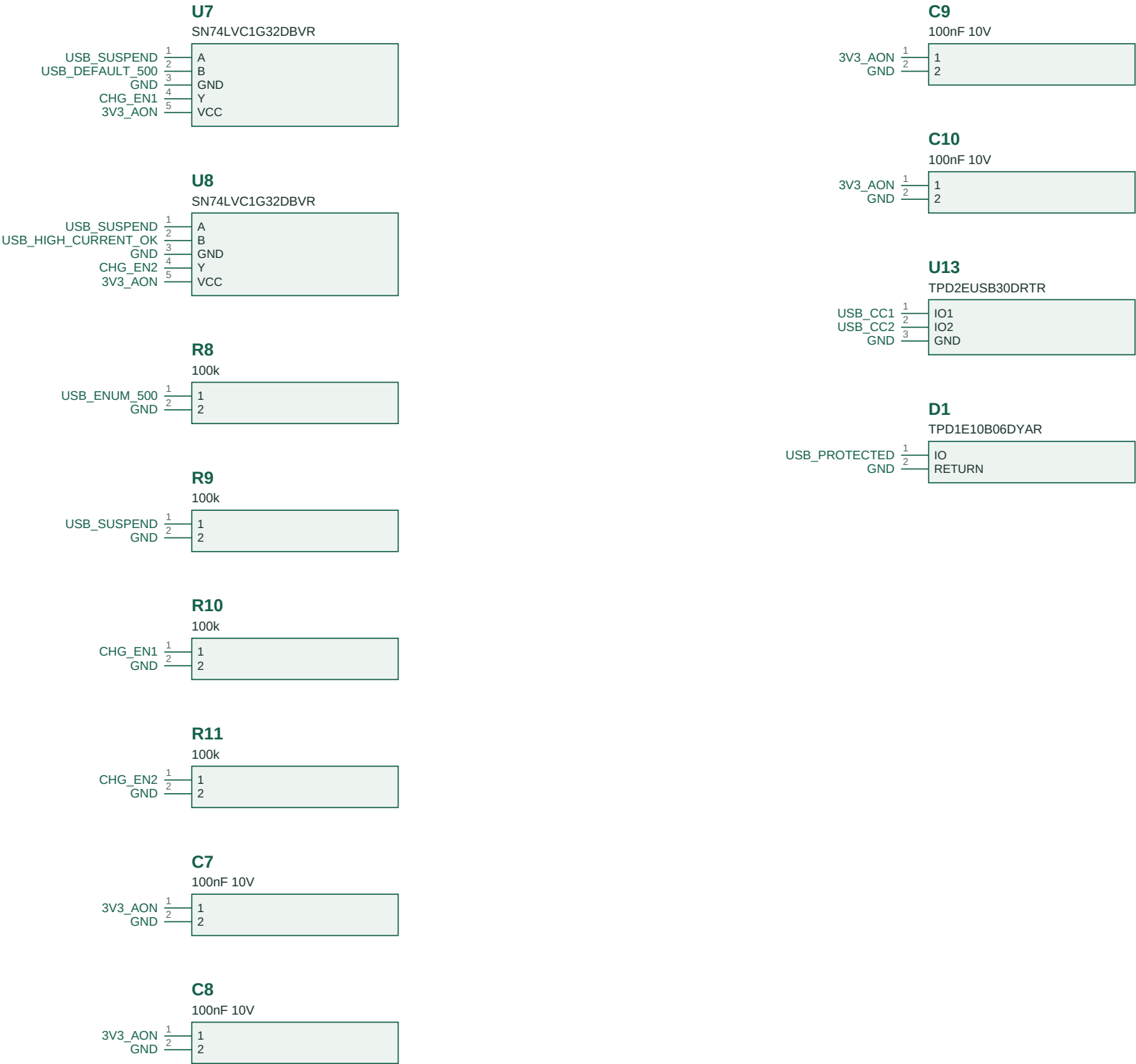
Circuit 01 / 01_usb_charge

Same global label = same electrical net. Pin numbers are physical package contacts. NC is intentional no-connect.



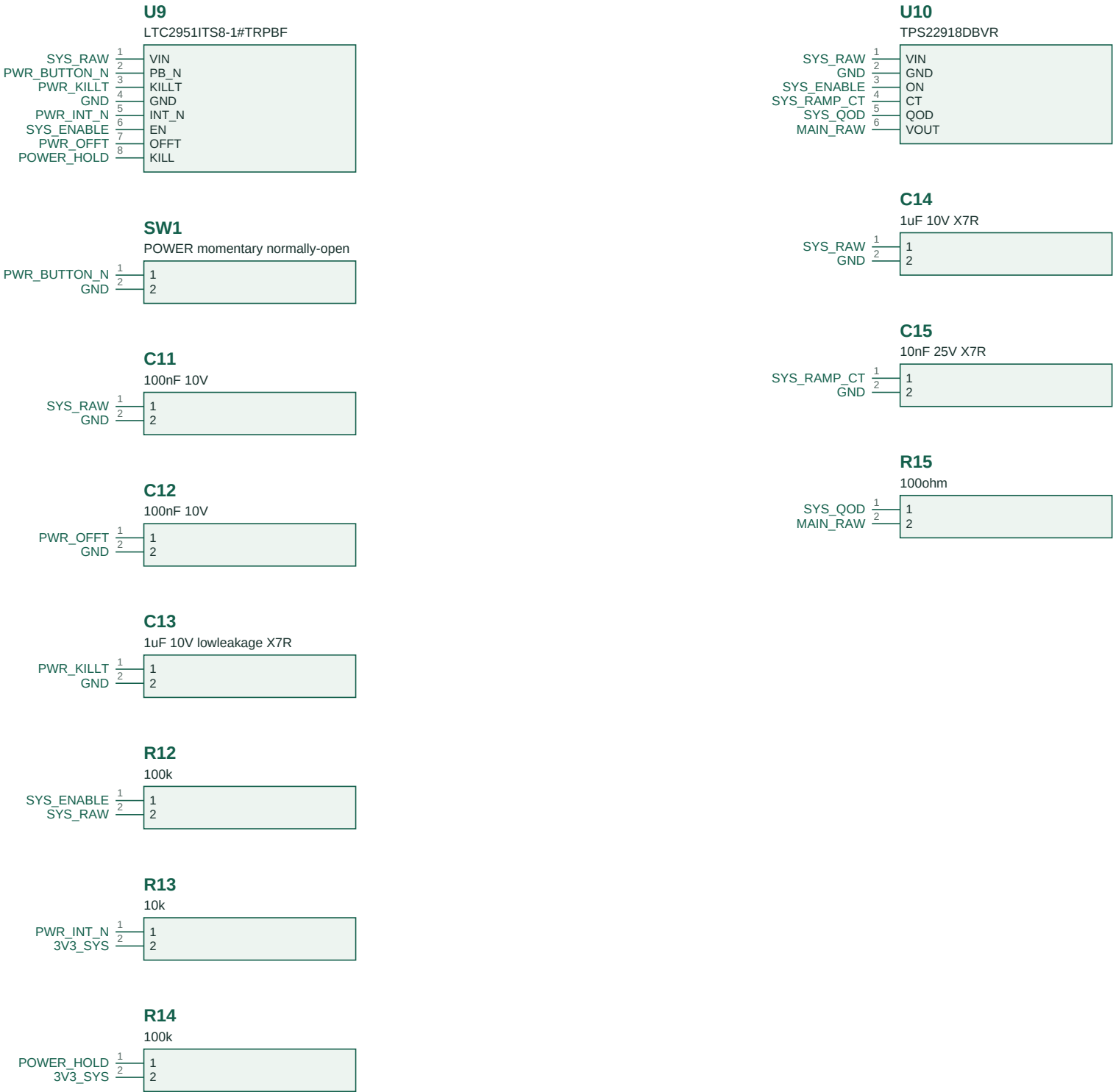
Circuit 02 / 01_usb_charge

Same global label = same electrical net. Pin numbers are physical package contacts. NC is intentional no-connect.



Circuit 03 / 02_power_control

Same global label = same electrical net. Pin numbers are physical package contacts. NC is intentional no-connect.



C12

100nF 10V

1

PWR_OFFT

2

GND

C13

1uF 10V lowleakage X7R

1

PWR_KILLT

2

GND

R12

100k

1

SYS_ENABLE

2

SYS_RAW

R13

10k

1

PWR_INT_N

2

3V3_SYS

R14

100k

1

POWER_HOLD

2

3V3_SYS

U10

TPS22918DBVR

1

SYS_RAW

2

GND

3

ON

4

CT

5

QOD

6

VOUT

C14

1uF 10V X7R

1

SYS_RAW

2

GND

C15

10nF 25V X7R

1

SYS_RAMP_CT

2

GND

R15

100ohm

1

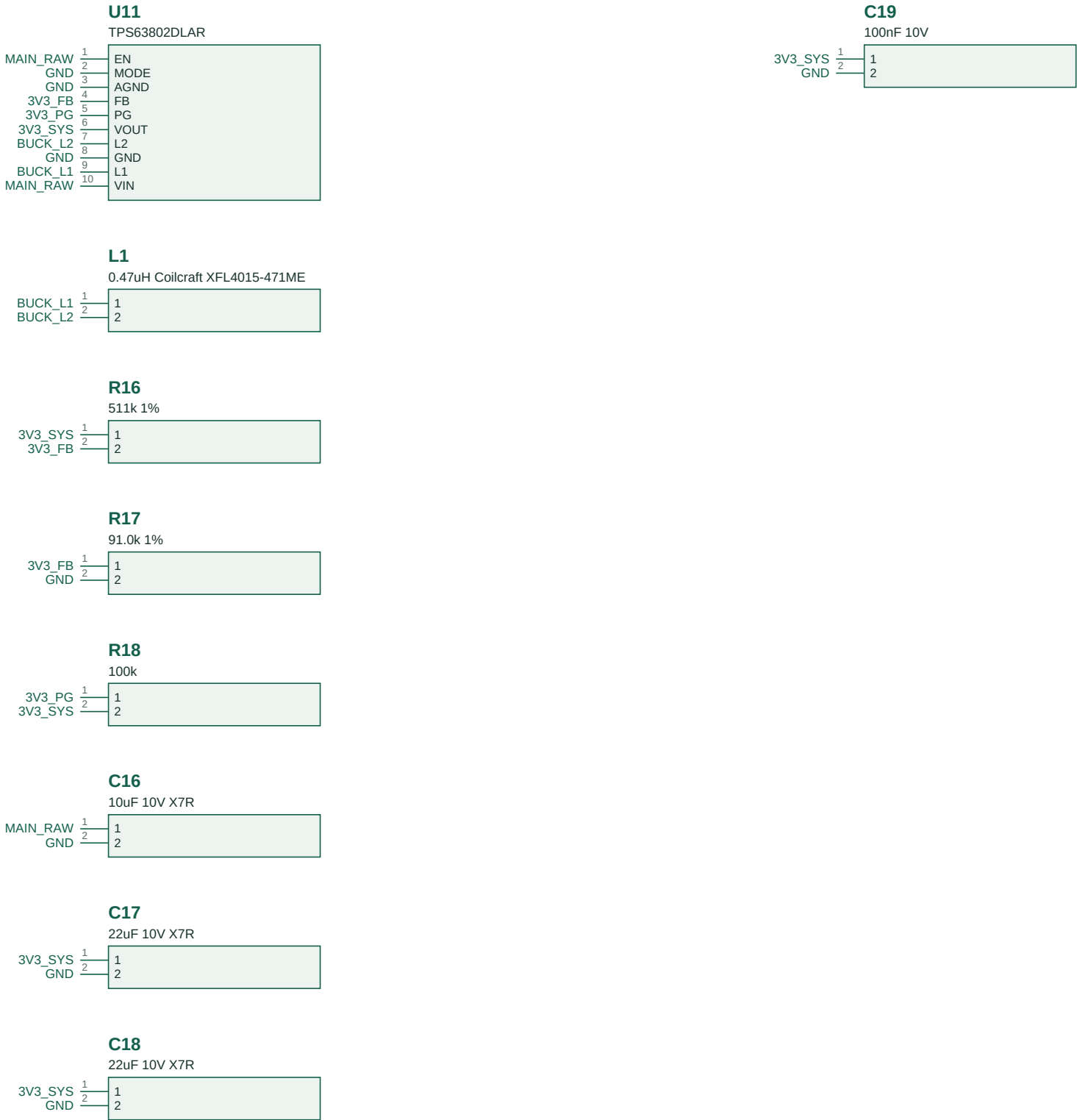
SYS_QOD

2

MAIN_RAW

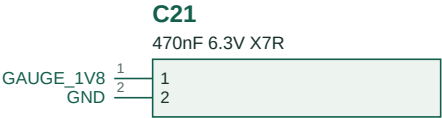
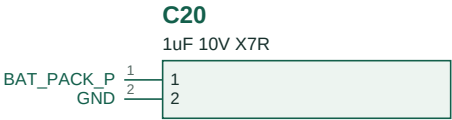
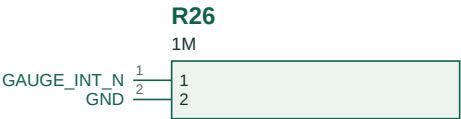
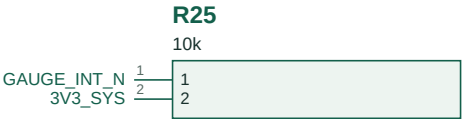
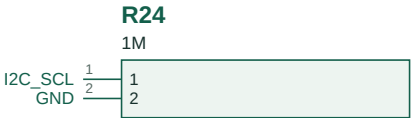
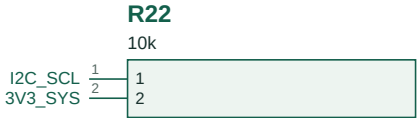
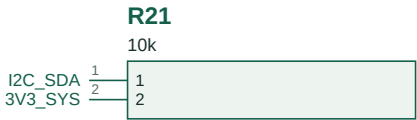
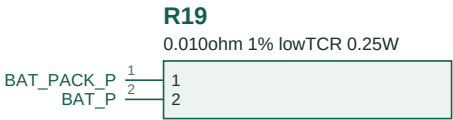
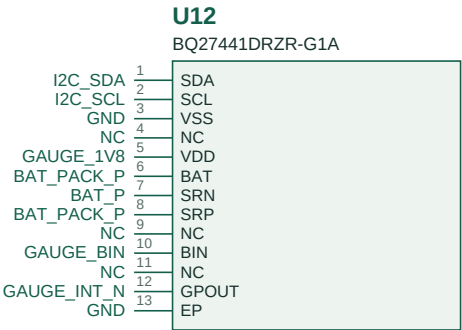
Circuit 04 / 03_regulator

Same global label = same electrical net. Pin numbers are physical package contacts. NC is intentional no-connect.



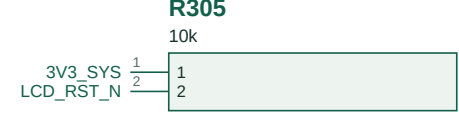
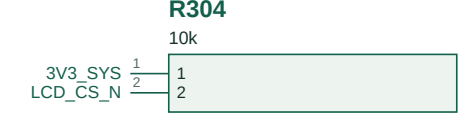
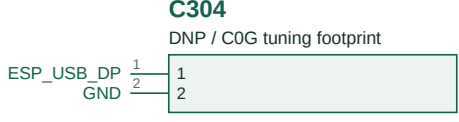
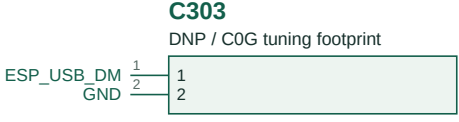
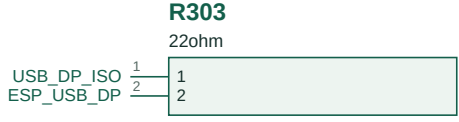
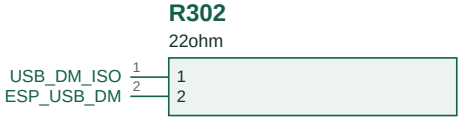
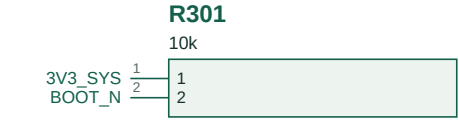
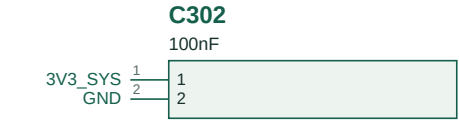
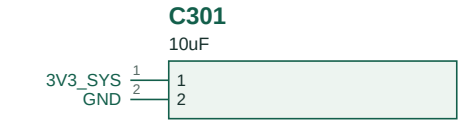
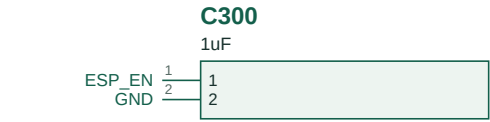
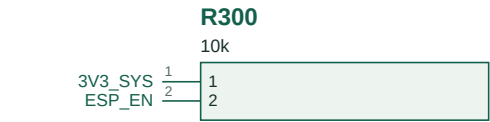
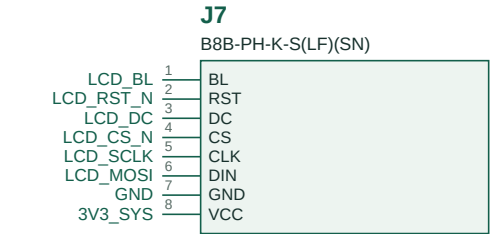
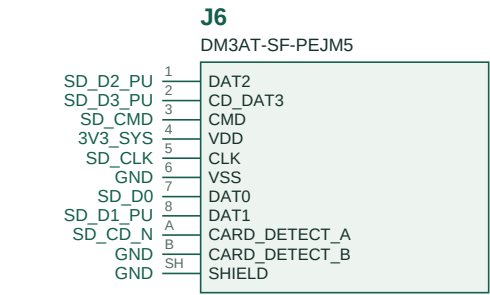
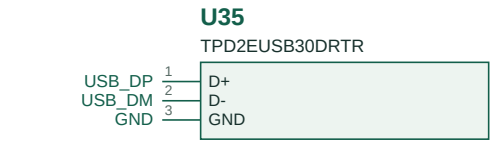
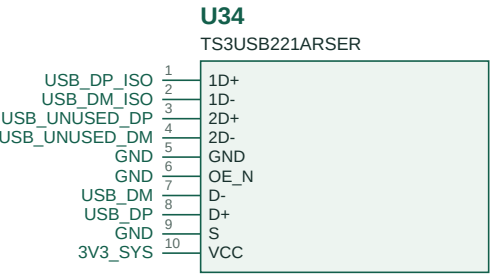
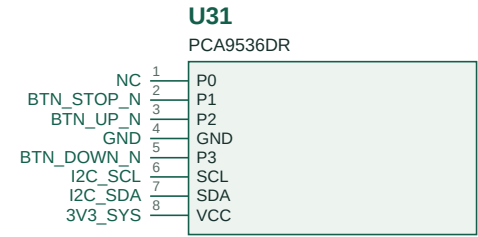
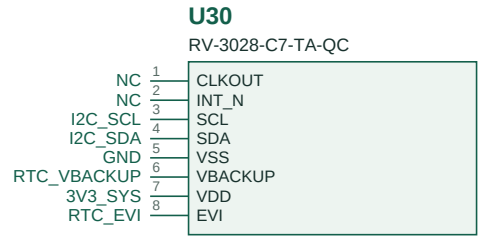
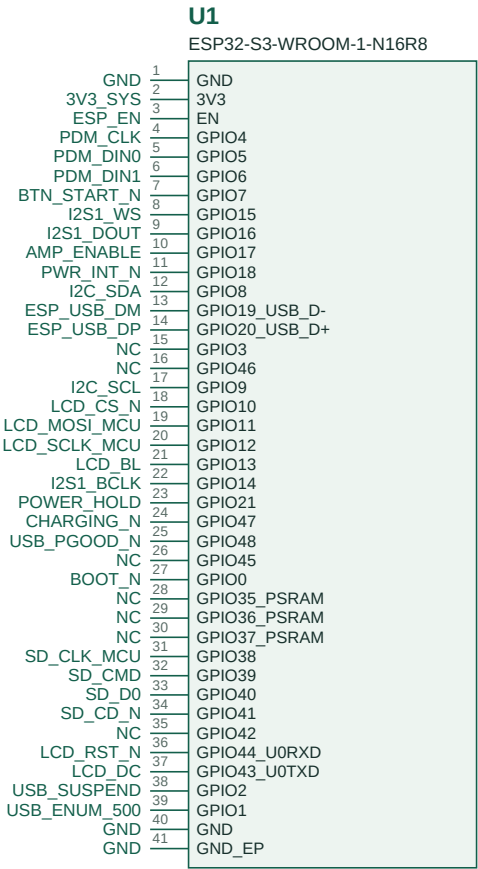
Circuit 05 / 04_gauge

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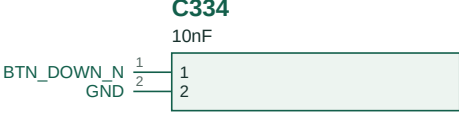
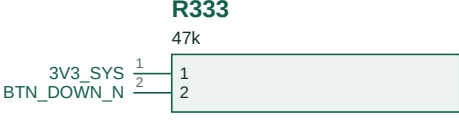
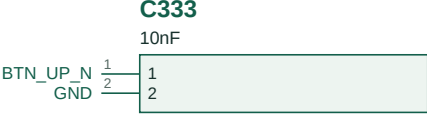
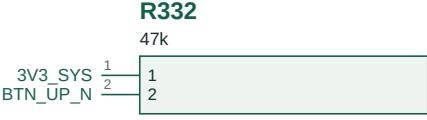
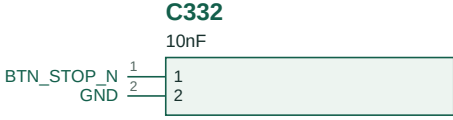
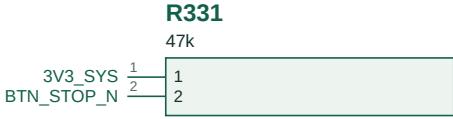
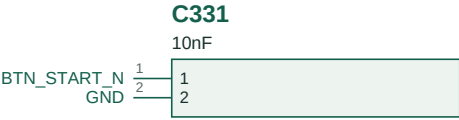
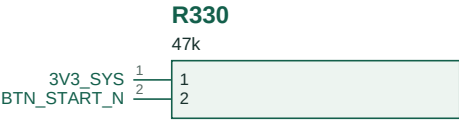
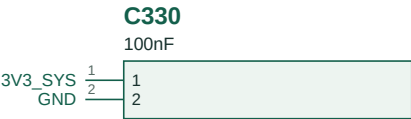
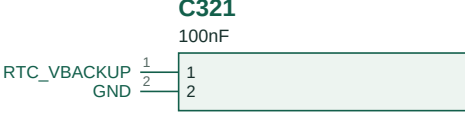
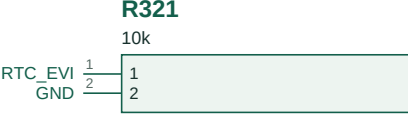
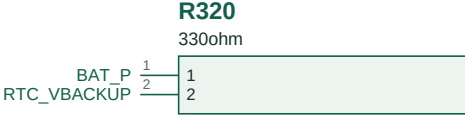
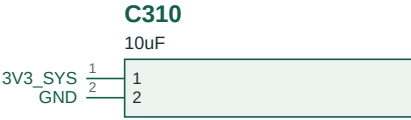
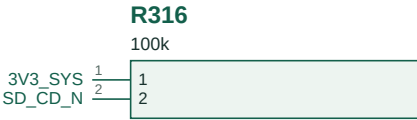
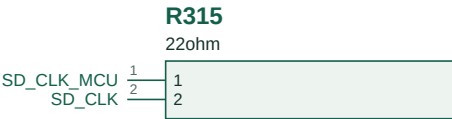
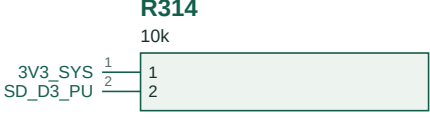
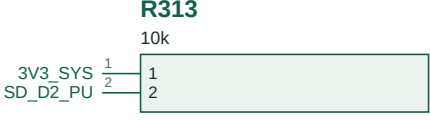
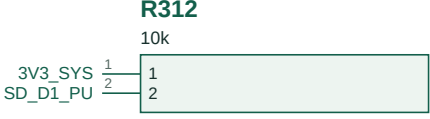
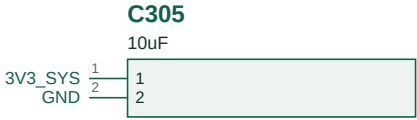
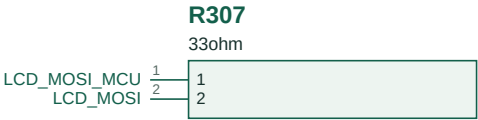
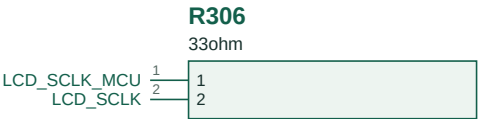
Circuit 06 / Controller

Same global label = same electrical net. Pin numbers are physical package contacts. NC is intentional no-connect.



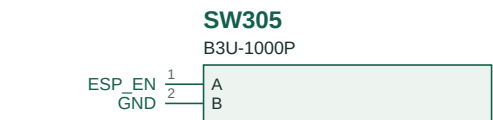
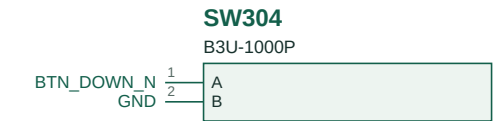
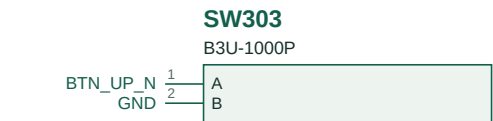
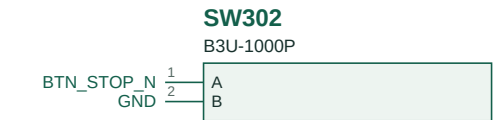
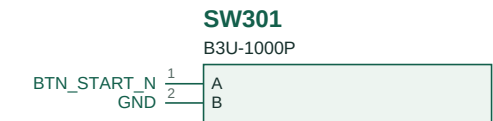
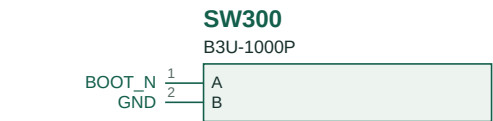
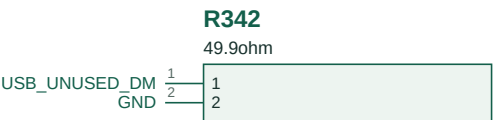
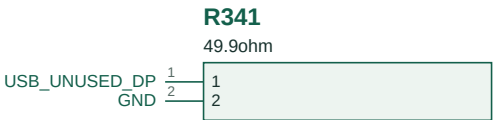
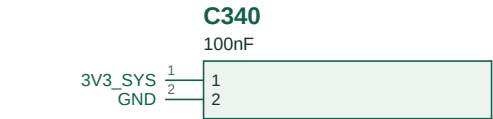
Circuit 07 / Controller

Same global label = same electrical net. Pin numbers are physical package contacts. NC is intentional no-connect.



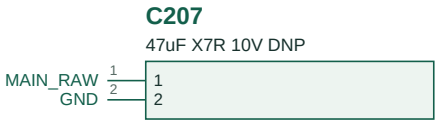
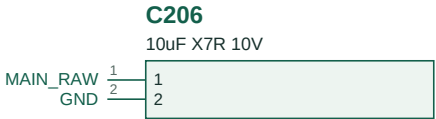
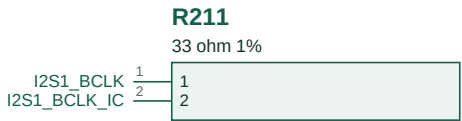
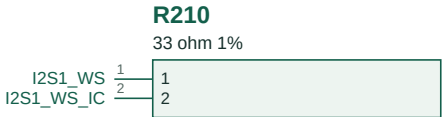
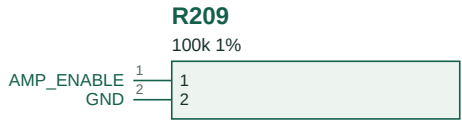
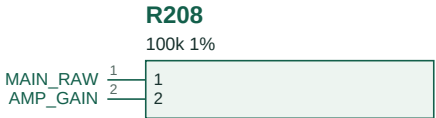
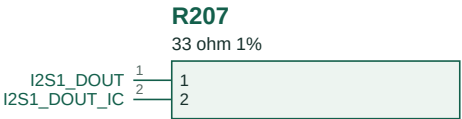
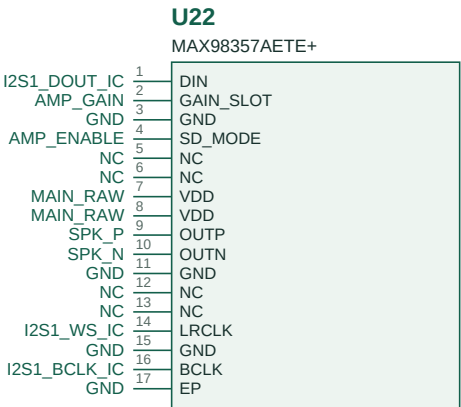
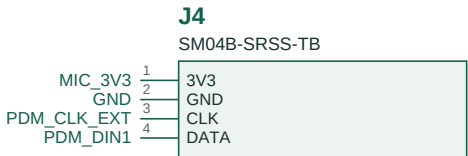
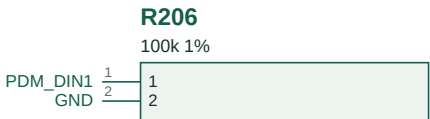
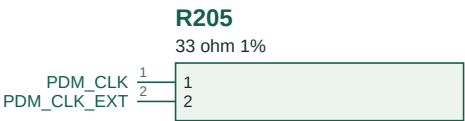
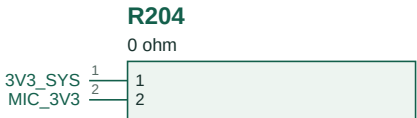
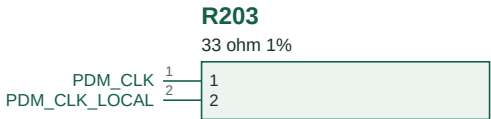
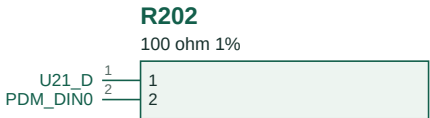
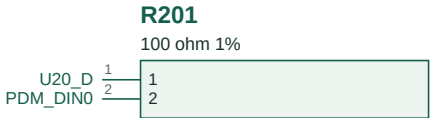
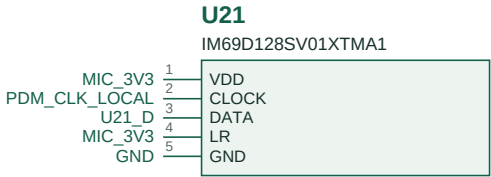
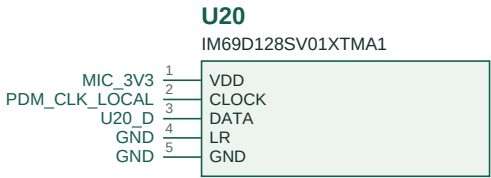
Circuit 08 / Controller

Same global label = same electrical net. Pin numbers are physical package contacts. NC is intentional no-connect.



Circuit 09 / Audio

Same global label = same electrical net. Pin numbers are physical package contacts. NC is intentional no-connect.



Manufacturer references

Circuit domain notes retain assumptions and selection details. Datasheets take precedence over drawings in this draft.

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